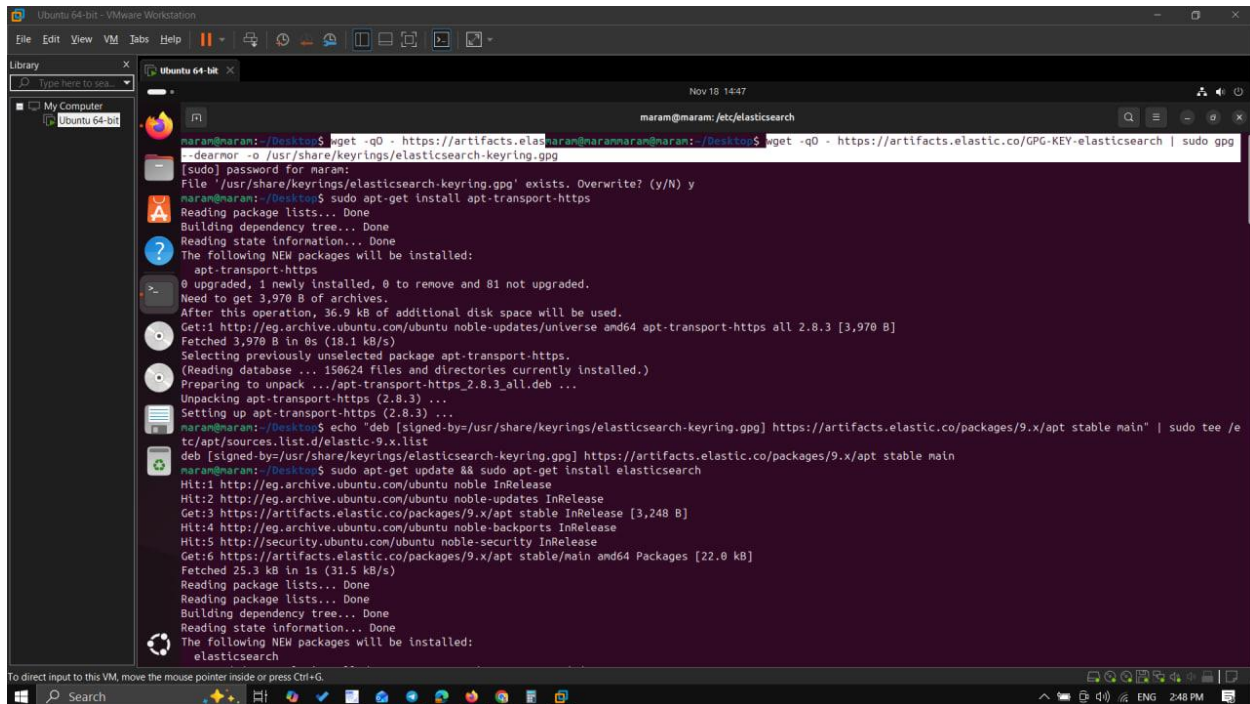


Install and configuration Elasticsearch and Kibana.

Downloading Elasticsearch

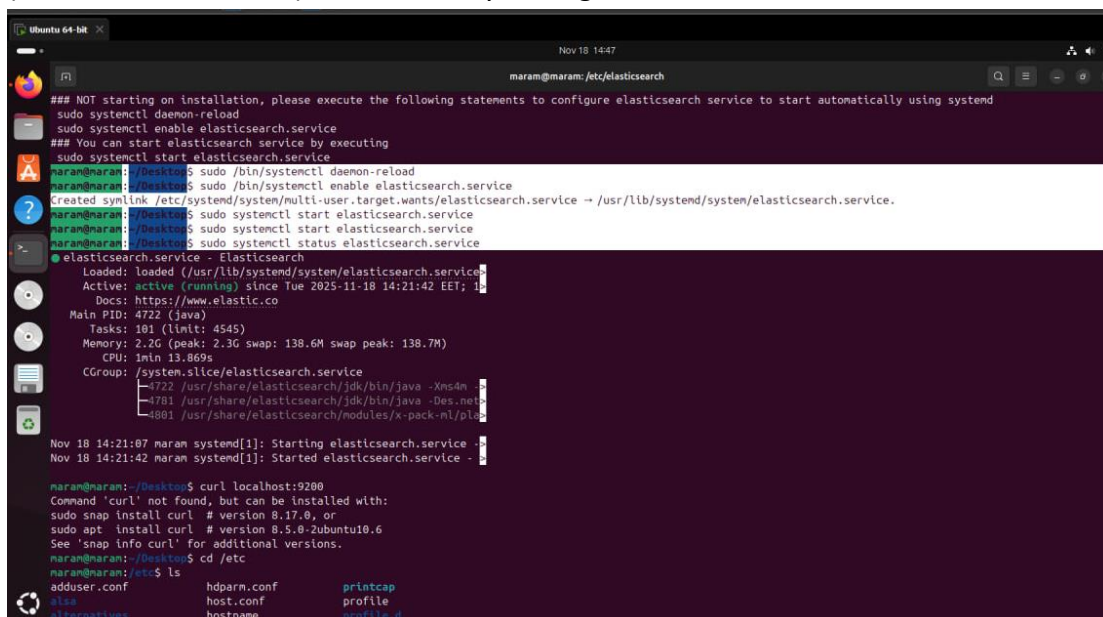
Visit the official Elastic website and download the appropriate Elasticsearch .tar.gz or .zip package for your operating system. This screenshot shows the `wget` command in action.



```
maram@maram: /etc/elasticsearch
maram@maram: /etc/elasticsearch$ wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg
--dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
[sudo] password for maram:
File '/usr/share/keyrings/elasticsearch-keyring.gpg' exists. Overwrite? (y/N) y
maram@maram: /etc/elasticsearch$ sudo apt-get install apt-transport-https
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  apt-transport-https
0 upgraded, 1 newly installed, 0 to remove and 81 not upgraded.
Need to get 3,970 B of archives.
After this operation, 36.9 kB of additional disk space will be used.
Get:1 http://eg.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3,970 B]
Fetched 3,970 B in 0s (18.1 kB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 158624 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Setting up apt-transport-https (2.8.3) ...
maram@maram: /etc/elasticsearch$ echo "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main" | sudo tee /e
tc/apt/sources.list.d/elasticsearch-9.x.list
deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main
maram@maram: /etc/elasticsearch$ sudo apt-get update && sudo apt-get install elasticsearch
Hit:1 http://eg.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eg.archive.ubuntu.com/ubuntu noble-updates InRelease
Get:3 https://artifacts.elastic.co/packages/9.x/apt stable InRelease [3,248 B]
Hit:4 http://eg.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://artifacts.elastic.co/packages/9.x/apt stable/main amd64 Packages [22.0 kB]
Fetched 25.3 kB in 1s (31.5 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  elasticsearch
```

Editing elasticsearch.yml Configuration

Open the `config/elasticsearch.yml` file to set cluster name, node name, network host (network.host: 0.0.0.0), and discovery settings.



```
maram@maram: /etc/elasticsearch
maram@maram: /etc/elasticsearch$ sudo systemctl daemon-reload
maram@maram: /etc/elasticsearch$ sudo systemctl enable elasticsearch.service
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /usr/lib/systemd/system/elasticsearch.service.
maram@maram: /etc/elasticsearch$ sudo systemctl start elasticsearch.service
maram@maram: /etc/elasticsearch$ sudo systemctl status elasticsearch.service
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; vendor preset: enabled)
   Active: active (running) since Tue 2025-11-18 14:21:42 EET; 1s ago
     Docs: https://www.elastic.co
   Main PID: 4722 (java)
    Tasks: 101 (limit: 4545)
   Memory: 2.2G (peak: 2.3G swap: 138.6M swap peak: 138.7M)
      CPU: 1min 13.869s
     CGroup: /system.slice/elasticsearch.service
            └─4722 /usr/share/elasticsearch/jdk/bin/java -Xms4m
              4781 /usr/share/elasticsearch/jdk/bin/java -Des.net
              4801 /usr/share/elasticsearch/modules/x-pack-ml/plu
Nov 18 14:21:07 maram systemd[1]: Starting elasticsearch.service:
Nov 18 14:21:42 maram systemd[1]: Started elasticsearch.service:
maram@maram: /etc$ curl localhost:9200
Command 'curl' not found, but can be installed with:
sudo snap install curl # version 8.17.0, or
sudo apt install curl # version 8.5.0-2ubuntu10.6
See 'snap info curl' for additional versions.
maram@maram: /etc$ cd /etc
maram@maram: /etc$ ls
adduser.conf      hdparm.conf      printcap
aliases           host.conf        profile
alternatives     hostname         profile.d
```

```
groff          passwd-          vulkan
group          pcnclia        wgetrc
group-         perl          wpa_supplicant
group-         perl          x11
gshadow        plymouth      xattr.conf
gshadow-       pm            xdg
gpg            pnm2ppa.conf xmi
gtk-2.0        polkit-1      zsh_command_not_found
gtk-3.0        ppp

maram@maram:/etc$ sudo chmod 777 elasticsearch/
maram@maram:/etc$ cd elasticsearch/
maram@maram:/etc/elasticsearch$ ls
certificates  jvm.options  roles.yml
elasticsearch.keystore  jvm.options.d  users
elasticsearch-plugins.example.yml  log4j2.properties  users_roles
elasticsearch.yml  role_mapping.yml

maram@maram:/etc/elasticsearch$ sudo chmod 777 elasticsearch.yml
maram@maram:/etc/elasticsearch$ sudo nano elasticsearch.yml
maram@maram:/etc/elasticsearch$ sudo systemctl restart elasticsearch.service
maram@maram:/etc/elasticsearch$ sudo systemctl status elasticsearch.service
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; vendor preset: enabled)
   Active: active (running) since Tue 2025-11-18 14:42:33 EET; 1min 10s ago
     Docs: https://www.elastic.co
    Main PID: 6220 (java)
      Tasks: 87 (limit: 4545)
     Memory: 1.9G (peak: 1.9G swap: 563.8M swap peak: 563.9M)
        CPU: 1min 9.867s
      CGroup: /system.slice/elasticsearch.service
              └─6220 /usr/share/elasticsearch/jdk/bin/java -Xms4g -Xmx4g -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.refreshinterval=30s -Des.jdk.options=
                  6281 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.refreshinterval=30s -Des.jdk.options=
                  6301 /usr/share/elasticsearch/modules/x-pack-ml/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.refreshinterval=30s -Des.jdk.options=

Nov 18 14:41:49 maram systemd[1]: Starting elasticsearch.service: Elasticsearch.
Nov 18 14:42:33 maram systemd[1]: Started elasticsearch.service: Elasticsearch.

maram@maram:/etc/elasticsearch$
```

Verifying Elasticsearch is Running

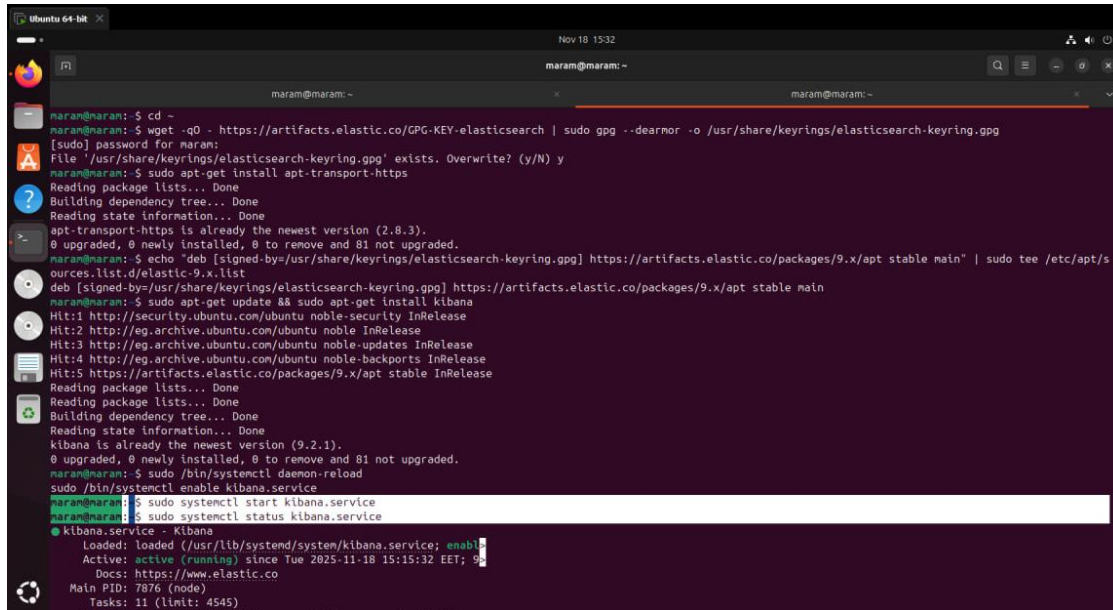
Access `http://localhost:9200` in a browser or using curl. The response should include cluster name, version number, and build hash.

```
{
  "name": "maram",
  "cluster_name": "elasticsearch",
  "cluster_uuid": "Km_F0R76Rj2kYveAEy_ooA",
  "version": {
    "number": "9.2.2",
    "build_flavor": "default",
    "build_type": "deb",
    "build_hash": "ed771e6976fac1a085affabd45433234a4babeaf",
    "build_date": "2025-11-27T08:06:51.614397514Z",
    "build_snapshot": false,
    "lucene_version": "10.3.2",
    "minimum_wire_compatibility_version": "8.19.0",
    "minimum_index_compatibility_version": "8.0.0"
  },
  "tagline": "You Know, for Search"
}
```

Downloading Kibana & Extracting and Configuring Kibana

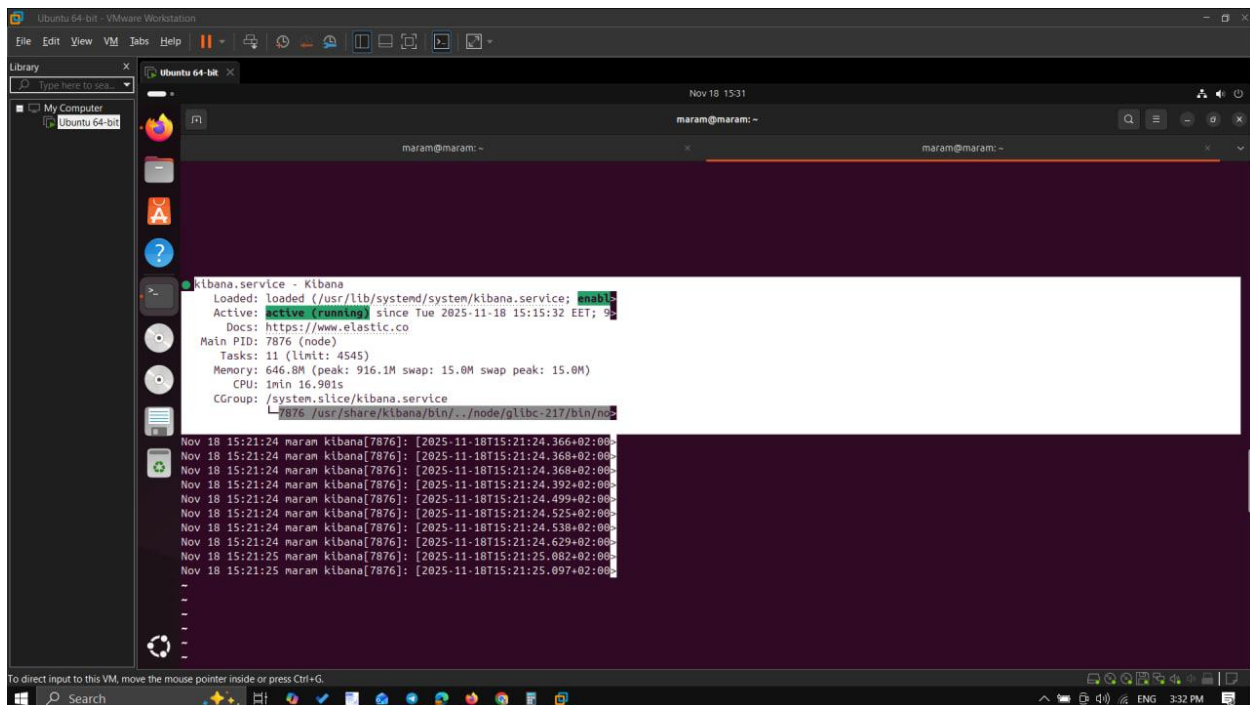
Download the Kibana package matching the Elasticsearch version from the official Elastic website. This screenshot show the wegt step.

Extract the Kibana archive and edit config/kibana.yml. Set elasticsearch.hosts: ["http://localhost:9200"] and server.host: "0.0.0.0".



```
maram@maram:~$ cd -
maram@maram:~$ wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
[sudo] password for maram:
File '/usr/share/keyrings/elasticsearch-keyring.gpg' exists. Overwrite? (y/N) y
maram@maram:~$ sudo apt-get install apt-transport-https
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apt-transport-https is already the newest version (2.8.3).
0 upgraded, 0 newly installed, 0 to remove and 81 not upgraded.
maram@maram:~$ sudo echo 'deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main' | sudo tee /etc/apt/sources.list.d/elastic-9.x.list
deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main
maram@maram:~$ sudo apt-get update && sudo apt-get install kibana
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://eg.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://eg.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://eg.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 https://artifacts.elastic.co/packages/9.x/apt stable InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
kibana is already the newest version (9.2.1).
0 upgraded, 0 newly installed, 0 to remove and 81 not upgraded.
maram@maram:~$ sudo /bin/systemctl daemon-reload
sudo /bin/systemctl enable kibana.service
maram@maram:~$ sudo systemctl start kibana.service
maram@maram:~$ sudo systemctl status kibana.service
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-11-18 15:15:32 EET; 9s
     Docs: https://www.elastic.co
   Main PID: 7876 (node)
    Tasks: 11 (limit: 4545)
```

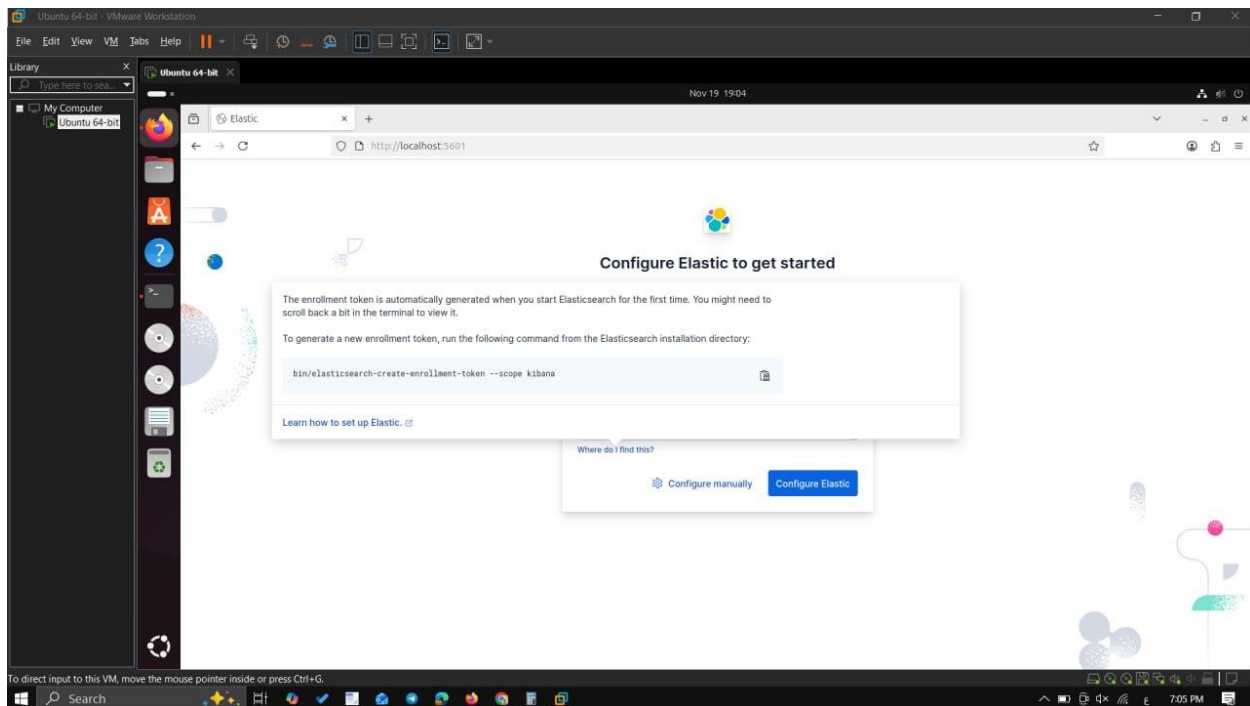
Starting Kibana



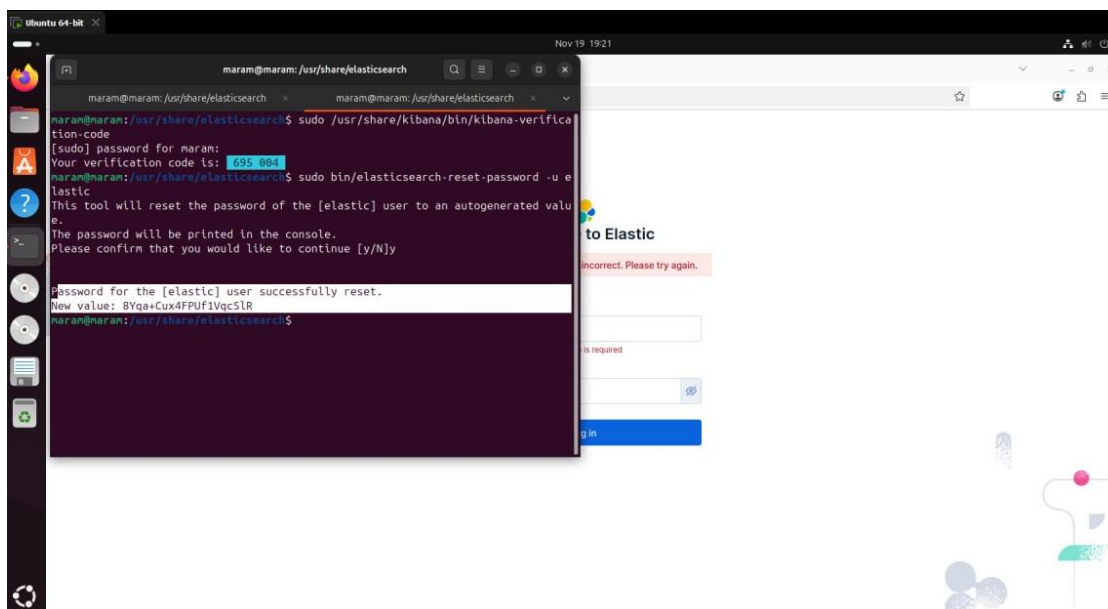
```
kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-11-18 15:15:32 EET; 9s
     Docs: https://www.elastic.co
   Main PID: 7876 (node)
    Tasks: 11 (limit: 4545)
   Memory: 646.8M (peak: 916.1M swap: 15.0M swap peak: 15.0M)
      CPU: 1min 16.901s
   CGroup: /system.slice/kibana.service
           └─7876 /usr/share/kibana/bin/node /usr/share/kibana/bin/node

Nov 18 15:21:24 maram kibana[7876]: [2025-11-18T15:21:24.366+02:00]
Nov 18 15:21:24 maram kibana[7876]: [2025-11-18T15:21:24.368+02:00]
Nov 18 15:21:24 maram kibana[7876]: [2025-11-18T15:21:24.368+02:00]
Nov 18 15:21:24 maram kibana[7876]: [2025-11-18T15:21:24.392+02:00]
Nov 18 15:21:24 maram kibana[7876]: [2025-11-18T15:21:24.499+02:00]
Nov 18 15:21:24 maram kibana[7876]: [2025-11-18T15:21:24.525+02:00]
Nov 18 15:21:24 maram kibana[7876]: [2025-11-18T15:21:24.538+02:00]
Nov 18 15:21:24 maram kibana[7876]: [2025-11-18T15:21:24.629+02:00]
Nov 18 15:21:25 maram kibana[7876]: [2025-11-18T15:21:25.082+02:00]
Nov 18 15:21:25 maram kibana[7876]: [2025-11-18T15:21:25.097+02:00]
```

Elasticsearch Enrollment Token Setup – This screen appears during Kibana configuration. Use the token generated by `elasticsearch-create-enrollment-token --scope kibana` to connect Kibana to Elasticsearch.



Kibana Verification Code & Elastic Password Reset – Run `kibana-verification-code` to get a 6-digit code, and `elasticsearch-reset-password -u elastic` to generate a new password for the elastic user.



Accessing Kibana Web Interface

Open `http://localhost:5601` in a browser. The screenshot shows the Kibana home page, confirming successful installation and configuration.

